

Overview of materials for Polybutene

Categories: [Polymer](#); [Thermoplastic](#); [Polybutene](#)

Material Notes: This property data is a summary of similar materials in the MatWeb database for the category "Polybutene". Each property range of values reported is minimum and maximum values of appropriate MatWeb entries. The comments report the average value, and number of data points used to calculate the average. The values are not necessarily typical of any specific grade, especially less common values and those that can be most affected by additives or processing methods.

Vendors: No vendors are listed for this material. Please [click here](#) if you are a supplier and would like information on how to add your listing to this material.

Physical Properties	Metric	English	Comments
Density	0.870 - 0.918 g/cc	0.0314 - 0.0332 lb/in ³	Average value: 0.902 g/cc Grade Count:36
Viscosity	3200 - 640000 cP @Temperature 117 - 190 °C	3200 - 640000 cP @Temperature 243 - 374 °F	Average value: 135000 cP Grade Count:7
Environmental Stress Crack Resistance	>= 15000 hour @Temperature 50.0 - 50.0 °C	>= 15000 hour @Temperature 122 - 122 °F	Average value: 15000 hour Grade Count:3
Melt Flow	0.400 - 2500 g/10 min	0.400 - 2500 g/10 min	Average value: 191 g/10 min Grade Count:36

Mechanical Properties	Metric	English	Comments
Tensile Strength, Ultimate	12.0 - 35.0 MPa	1740 - 5080 psi	Average value: 27.2 MPa Grade Count:22
Film Tensile Strength at Yield, MD	15.0 - 23.0 MPa	2180 - 3340 psi	Average value: 19.0 MPa Grade Count:3
Tensile Strength, Yield	3.74 - 17.2 MPa	542 - 2490 psi	Average value: 10.6 MPa Grade Count:8
Film Elongation at Break, MD	580 - 790 %	580 - 790 %	Average value: 708 % Grade Count:5
Elongation at Break	300 - 800 %	300 - 800 %	Average value: 343 % Grade Count:21
Modulus of Elasticity	0.0517 - 0.293 GPa	7.50 - 42.5 ksi	Average value: 0.163 GPa Grade Count:10
Flexural Modulus	0.0100 - 0.450 GPa	1.45 - 65.3 ksi	Average value: 0.240 GPa Grade Count:19
Secant Modulus, MD	0.185 - 0.550 GPa	26.8 - 79.8 ksi	Average value: 0.315 GPa Grade Count:5
Film Tensile Strength at Break, MD	28.0 - 34.0 MPa	4060 - 4930 psi	Average value: 31.0 MPa Grade Count:5

Thermal Properties	Metric	English	Comments
Melting Point	81.0 - 128 °C	178 - 262 °F	Average value: 106 °C Grade Count:25
Vicat Softening Point	90.0 - 113 °C	194 - 235 °F	Average value: 102 °C Grade Count:8
Brittleness Temperature	-34.0 - -17.0 °C	-29.2 - 1.40 °F	Average value: -22.7 °C Grade Count:7

Some of the values displayed above may have been converted from their original units and/or rounded in order to display the information in a consistent format. Users requiring more precise data for scientific or engineering calculations can click on the property value to see the original value as well as raw conversions to equivalent units. We advise that you only use the original value or one of its raw conversions in your calculations to minimize rounding error. We also ask that you refer to MatWeb's [terms of use](#) regarding this information. [Click here](#) to view all the property values for this datasheet as they were originally entered into MatWeb.